

Neutrino Oscillation

In vacuum:

$$P_{ee} = 1 - \sin^2 2\theta \sin^2 \frac{\Delta m^2 L}{4E}$$

$$P_{e\mu} = \sin^2 2\theta \sin^2 \frac{\Delta m^2 L}{4E}$$

Practically useful:

$$P_{e\mu} = \sin^2 2\theta \sin^2 \left(1.27 \frac{\Delta m^2}{10^{-3} \text{ eV}^2} \cdot \frac{L}{\text{km}} \cdot \frac{\text{MeV}}{E} \right)$$

Neutrino as Quantum Chameleon

Two mass-eigenstate phases

ν_1
 ν_2

in phase

out of phase

Oscillation probabilities

P_{ee}

$P_{e\mu}$

